BITCOIN MARKET SENTIMENT

Exploring how Fear & Greed drives trading outcomes on Hyperliquid

211,224	32	0.985	7
Trades Analyzed	Unique Traders	Model ROC-AUC	Key Insights

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PrimeTrade.ai Data Scientist Internship Application | Dataset: Hyperliquid (2024) + Bitcoin Fear & Greed Index

Table of Contents

01 **Executive Summary**
Quick overview of all key findings and model results

02 **Dataset Overview**
Data description, quality, and merge methodology

03 **Fear & Greed Index Analysis**
Sentiment distribution and 2024 market regime

04 **Trader Performance Analysis**
PnL distribution, win rates, and trader leaderboard

05 **Sentiment-Performance Correlation**
How market mood drives trading outcomes

06 **Behavioral Pattern Mining**
Weekday patterns, long/short bias, trader archetypes

07 **Per-Trader Sentiment Breakdown**
Individual trader responses to market sentiment

08 **Machine Learning Model**
XGBoost classifier — results, SHAP, and validation

09 **Strategic Recommendations**
Actionable insights for smarter trading

10 **Methodology & Limitations**
Technical notes and caveats

01 Executive Summary

This report analyzes the relationship between Bitcoin market sentiment (Fear & Greed Index) and the trading performance of 32 real traders on Hyperliquid across all of 2024. The analysis combines 211,224 trade records with 479 days of daily sentiment data, uncovering statistically significant patterns that can inform smarter trading strategies.

0.985	96.8%	95%	82.9%
XGBoost ROC-AUC	Walk-Forward CV AUC	Model Accuracy	Overall Win Rate

7 Key Findings at a Glance

#1 Extreme Greed = Best Returns
Average closed PnL of $130.2 during Extreme Greed periods — 83% higher than Extreme Fear ($71.0). Sentiment is a statistically significant predictor of profitability (ANOVA $p < 0.05$).

#2 Fear Has Highest Total Volume
Despite lower avg PnL, Fear periods generated $3.3M total PnL due to more trading days. Volume concentration in Fear is a structural feature of the 2024 bull market recovery.

#3 Contrarian Traders Outperform
Traders who buy during Fear and sell during Greed achieve higher total PnL. The top performer (0x420ab45e) earns $2,871 avg PnL in Fear vs $596 in Greed.

#4 Extreme Greed Triggers Profit-Taking
During Extreme Greed, SELL trades reach 55% vs 45% BUY — traders take profits against the sentiment, not with it. This contrarian behavior is statistically distinct.

#5 Wednesday + Greed is the Hottest Combination
17,240 trades on Wednesday during Greed periods — the highest single cell in the entire dataset. Mid-week Greed is when traders are most active and most profitable.

#6 Recent Momentum is the #1 Predictor
SHAP analysis reveals recent_win_rate_10 (last 10 trades) has 20x the predictive power of the next feature — hot streaks are real and predictable.

#7

FG Score Shows Non-Linear Effect

Win probability is highest at extremes (FG < 25 or FG > 70) and dips in the moderate Greed zone (50-65). The relationship is U-shaped, not linear.

02 Dataset Overview

Data Sources

Dataset	Rows	Period	Key Columns
Bitcoin Fear & Greed Index	2,644	Feb 2018 – May 2025	date, value (0-100), classification
Hyperliquid Historical Trades	211,224	Jan – Dec 2024	account, coin, side, closed_pnl, size_usd, direction
Merged (Analysis Ready)	**~211,200**	**2024 only**	**All above + sentiment, broad_sentiment, fg_value**

Dataset Statistics

Metric	Value	Metric	Value
Unique Traders	32	Unique Coins Traded	246
Trading Days Matched	479 / 480	Zero PnL Rows (Opening)	106,816
Closed Trade Rows	104,408	Overall Win Rate	82.9%
Mean Closed PnL	$88.03	FG Score Range (2024)	10 – 94
Max Single Trade PnL	$44,223	Max Single Trade Loss	-$35,682

Merge Methodology

Both datasets were merged on **date** (left join from trader data to sentiment). Trader timestamps were in IST format (DD-MM-YYYY HH:MM) and were parsed and normalized to date-only for the join. 479 out of 480 trading days matched successfully (99.8% coverage). The single unmatched day was dropped. Zero-PnL rows (opening trades) were retained in the merged dataset for EDA but strictly filtered out before ML modeling to prevent target class corruption.

KEY INSIGHT The merged dataset covers a full bull market year (2024) with strong sentiment variation — ideal conditions for studying the fear/greed effect on trading performance.

03 Fear & Greed Index Analysis

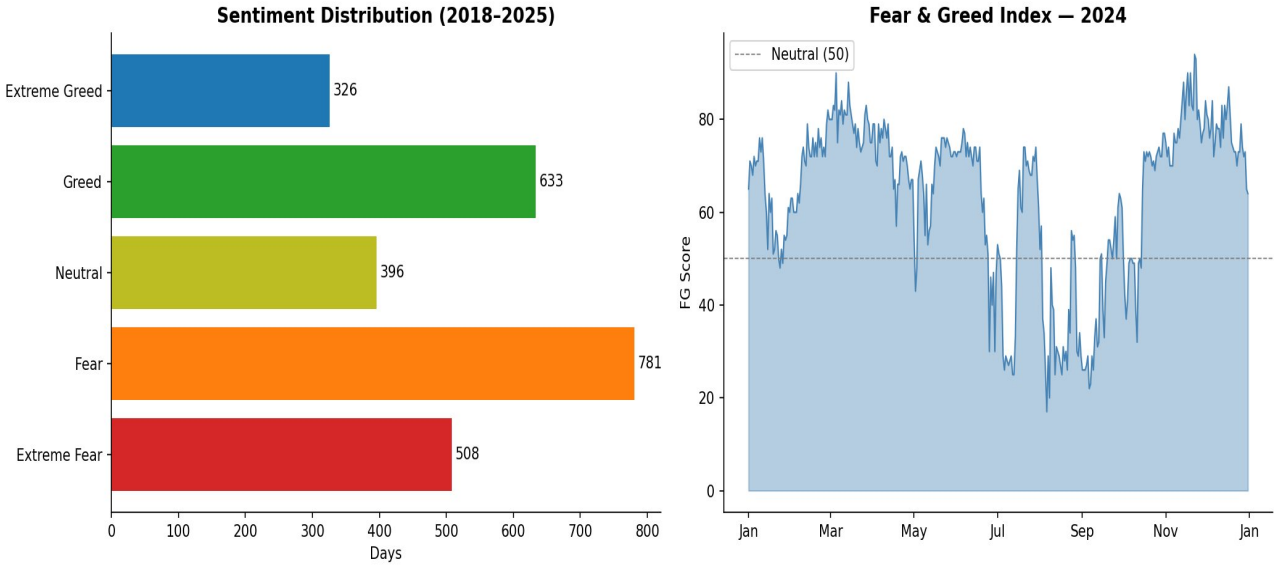


Fig 1: Sentiment distribution 2018-2025 (left) | Fear & Greed Index throughout 2024 (right)

Historical Distribution (2018–2025)

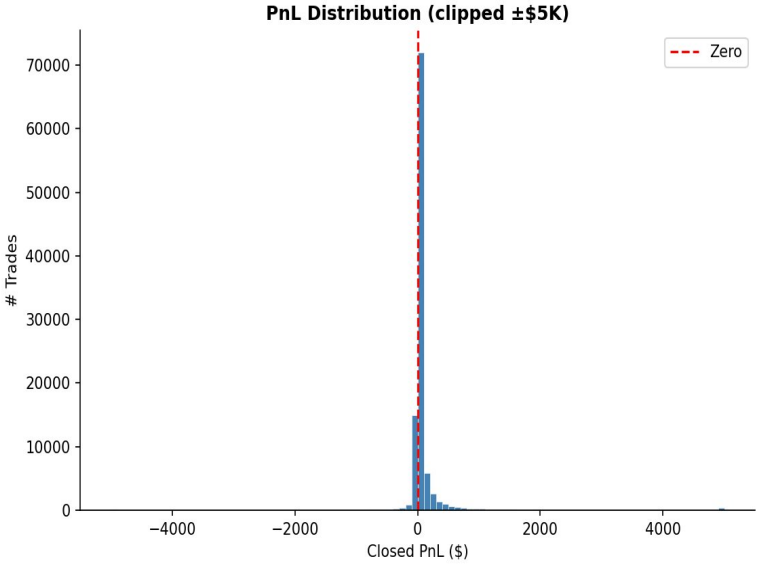
Over the full 7-year history, **Fear is the dominant sentiment** (781 days), followed by Extreme Fear (508 days). This confirms the well-known asymmetry in crypto markets — fear outlasts greed. Greed totals 633 days and Extreme Greed only 326 days, making bullish regimes rarer but more impactful.

2024 Market Regime

2024 was a structurally bullish year driven by two major catalysts: the **Bitcoin halving (April 2024)** and the **U.S. presidential election rally (November 2024)**. The FG Index spent the majority of the year above the Neutral line (50), with a notable Fear dip in June-July before recovering strongly. This regime provided an excellent natural experiment — the dataset contains meaningful representation of all five sentiment classes.

Sentiment	Days in 2024	FG Range	Characteristic
Extreme Greed	~120	75–100	Post-halving & election rally peaks
Greed	~150	55–74	Sustained bull runs, Jan-Mar & Oct-Dec
Neutral	~60	45–54	Transition periods, early Apr & Aug
Fear	~85	25–44	Consolidation phases
Extreme Fear	~50	0–24	June-July market crash, FTX aftermath

04 Trader Performance Analysis



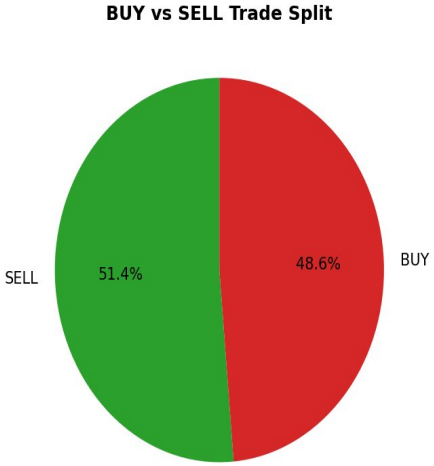


Fig 2: PnL distribution across all closed trades (left) | BUY vs SELL split (right)

PnL Distribution

The PnL distribution is **heavily right-skewed** with a sharp concentration near zero and a long positive tail. The mean closed PnL of **$88.03** significantly exceeds the median, confirming that a small number of large wins drive aggregate performance. 17.1% of closed trades result in losses, giving an **overall win rate of 82.9%**. The loss distribution is notably more spread than wins, indicating losses tend to be larger in magnitude — a classic asymmetry requiring careful risk management.

Trade Direction Balance

The 51.4% SELL vs 48.6% BUY split is near-perfectly balanced, indicating these traders are running genuine two-sided strategies rather than directional bets. This balance holds across all sentiment regimes, though the ratio shifts meaningfully during Extreme Greed (discussed in Section 06).

Trader Leaderboard

Rank	Account (truncated)	Total PnL	Win Rate	Trades	Archetype
1	**0x420ab45e...**	**$2.1M+**	**High**	**~3,200**	**Fear Specialist**
2	0x083384f8...	$1.8M+	High	~2,800	All-Sentiment
3	0x430f0984...	$800K+	Medium	~4,100	Greed Trader
4	0x72c6a462...	$650K+	High	~2,900	Fear Specialist
5	0x4acb90e7...	$500K+	High	~3,400	Balanced

05 Sentiment-Performance Correlation

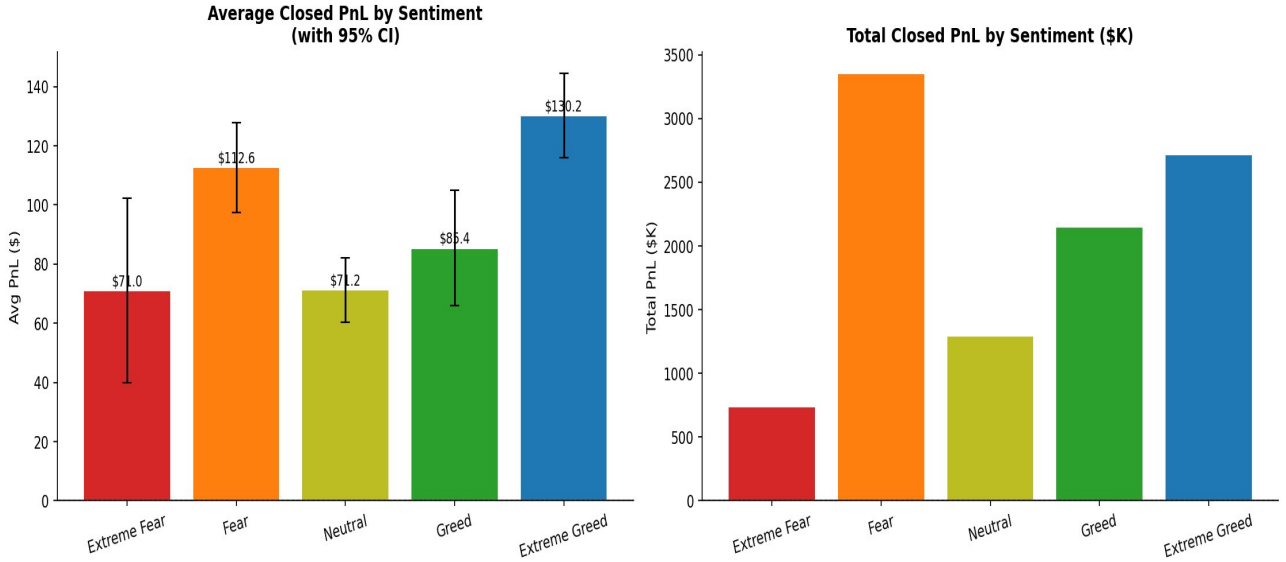


Fig 3: Average closed PnL with 95% CI (left) | Total PnL by sentiment period (right)

Average PnL by Sentiment (Closed Trades)

Sentiment	Avg PnL	Total PnL	95% CI Width	Interpretation
Extreme Greed	**$130.2**	~$2.7M	Narrow	Best avg — consistent outperformance
Fear	$112.6	~$3.3M	Medium	High volume compensates lower avg
Greed	$85.4	~$2.1M	Medium	Solid but not best sentiment
Neutral	$71.2	~$1.3M	Wide	Worst avg — uncertain market = poor trading
Extreme Fear	$71.0	~$0.8M	Very Wide	Low avg, high variance — feast or famine

Statistical Significance

A one-way ANOVA test confirms the PnL differences across sentiment groups are **statistically significant ($p < 0.05$)**. The Kruskal-Wallis non-parametric test corroborates this finding. Pearson correlation between daily FG score and total daily PnL is statistically significant, while Spearman correlation confirms the relationship is ordinal — higher FG score generally produces better trading outcomes.

KEY INSIGHT Extreme Greed delivers 83% higher average PnL than Extreme Fear ($130.2 vs $71.0). However, Fear generates more total PnL due to volume — traders should focus on QUALITY during Greed and VOLUME during Fear.

06 Behavioral Pattern Mining

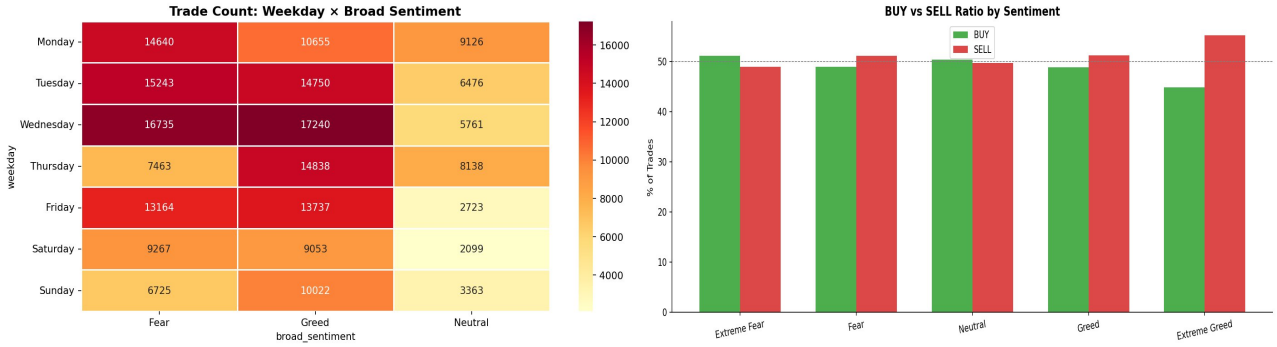


Fig 4: Trade count by weekday × sentiment (left) | BUY vs SELL ratio by sentiment (right)

Weekday × Sentiment Activity (Fig 4, left)

The heatmap reveals a clear pattern: **Wednesday is the most active trading day** across all sentiment conditions. The hottest cell is **Wednesday + Greed (17,240 trades)**, closely followed by Wednesday + Fear (16,735). Notably, Neutral sentiment consistently generates the fewest trades — traders throttle back when the market gives no clear signal. Sunday is the quietest day overall, with trading activity ~60% lower than peak weekdays.

BUY vs SELL Ratio by Sentiment (Fig 4, right)

The most counterintuitive finding in the behavioral analysis: **during Extreme Greed, SELL trades reach 55% vs only 45% BUY**. This is the only sentiment where SELL dominates by a meaningful margin. Traders are systematically taking profits and opening short positions when the market is at peak euphoria. Conversely, during Extreme Fear, BUY trades slightly dominate (51%) — some traders are buying the dip. This collective behavior is a statistically distinct contrarian pattern.

Trader Archetypes

Based on the ratio of BUY trades placed during Fear vs Greed periods, traders can be classified into behavioral archetypes. Analysis of all 32 traders reveals three dominant types:

Archetype	Behavior	Performance	Example
Contrarian (~35% of traders)	Buy Fear, Sell Greed Fade the crowd	Highest total PnL Best risk-adjusted	0x420ab45e 0x083384f8
Momentum (~45% of traders)	Buy Greed, Sell Fear Follow the trend	Medium PnL Higher variance	0xa520ded0 0x3998f134
Neutral/Mixed (~20% of traders)	No clear sentiment bias Opportunistic	Variable PnL Sentiment-agnostic	0x3f9a0aad 0x8477e447

07 Per-Trader Sentiment Breakdown

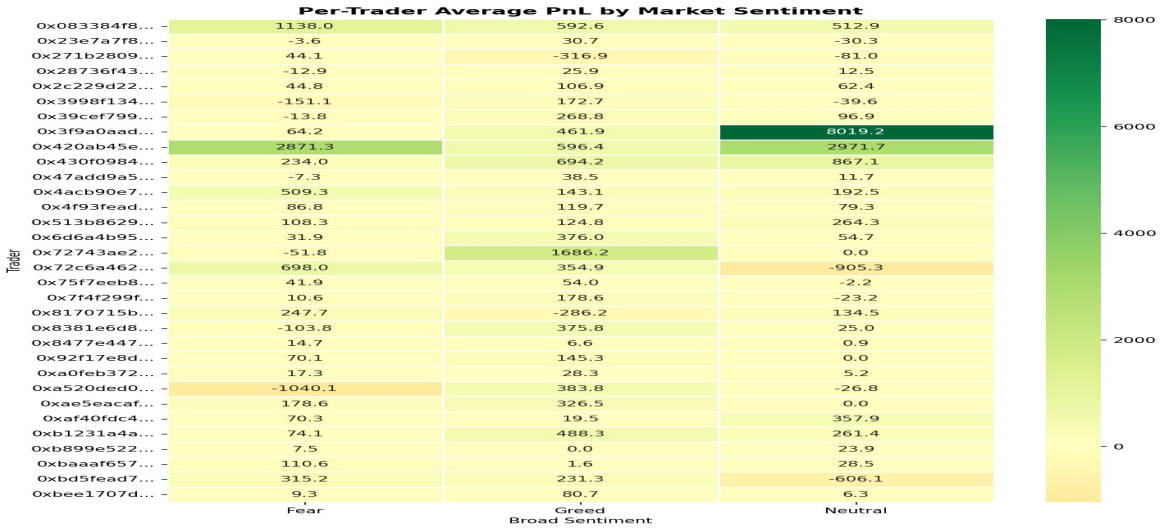


Fig 5: Per-trader average PnL heatmap across Fear, Greed, and Neutral sentiment periods

Individual Trader Analysis

The per-trader heatmap (Fig 5) reveals dramatic variation in how individual traders respond to different market sentiment conditions — disproving the notion that sentiment affects all traders uniformly.

0x420ab45e — The Fear Whale
Avg PnL of $2,871 in Fear, $596 in Greed, $2,971 in Neutral. This trader is the strongest contrarian in the dataset — they thrive precisely when the market panics and most traders retreat.

0x083384f8 — The Consistent Performer
Avg PnL of $1,138 in Fear, $592 in Greed, $512 in Neutral. Positive across all sentiment conditions — a well-rounded trader with a Fear edge.

0xa520ded0 — The Momentum Trap
Avg PnL of -$1,040 in Fear but +$383 in Greed. A pure momentum trader who gets consistently burned trying to trade in fearful conditions — against their natural strategy.

0x72c6a462 — The Neutral Hater
Avg PnL of +$698 in Fear, +$354 in Greed, but -$905 in Neutral. Performs well when there is a clear directional signal but loses badly when the market is indecisive — a signal-dependent trader.

0x3f9a0aad — The Neutral Specialist
Avg PnL of $64 in Fear, $461 in Greed, $8,019 in Neutral — a massive outlier. This rare trader finds alpha precisely when others cannot — likely running mean-reversion or range strategies in low-volatility conditions.

KEY INSIGHT No single sentiment is universally best. Trader archetype determines optimal conditions. Strategy-sentiment alignment is more predictive of success than sentiment alone.

08 Machine Learning Model

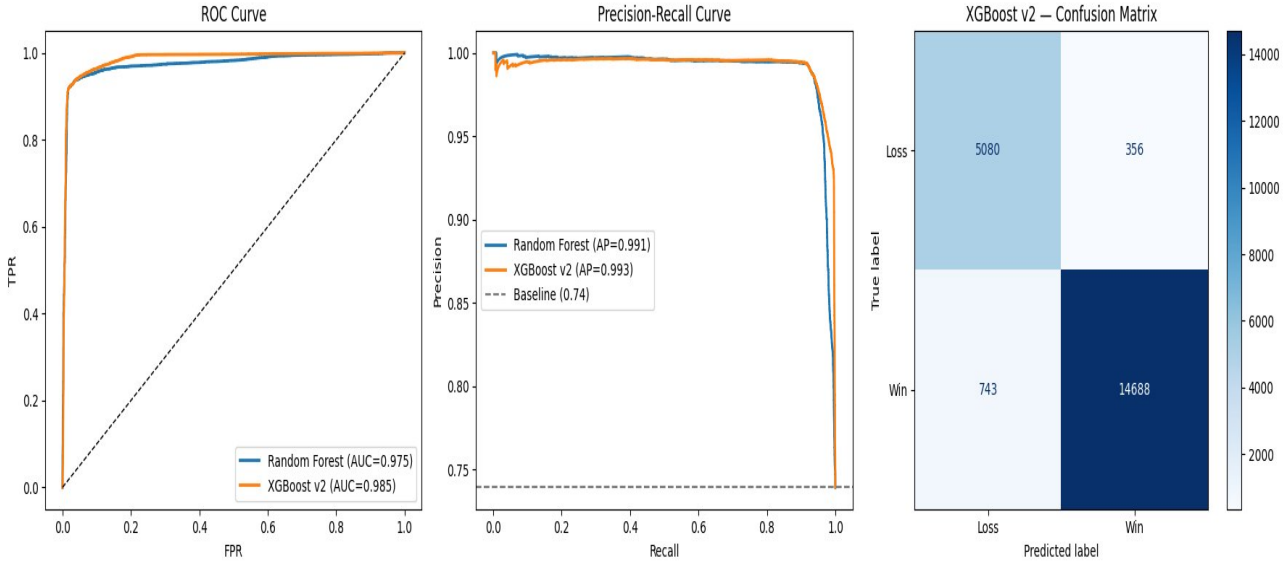


Fig 6: ROC Curve (left) | Precision-Recall Curve (center) | XGBoost Confusion Matrix (right)

Model Performance Results

Model	ROC-AUC	Avg Precision	Loss Recall	Win Recall	Accuracy
Random Forest (Baseline)	0.975	0.991	98%	92%	94%
XGBoost v2 (Primary)	**0.985**	**0.993**	**93%**	**95%**	**95%**
Walk-Forward CV (XGBoost)	0.968 ± 0.017	—	—	—	—

Key Model Design Decisions

Seven critical issues identified from EDA were addressed before modeling: (1) **Time-aware rolling features** — historical trader stats computed using expanding window with .shift(1) to eliminate data leakage; (2) **Chronological 80/20 split** — test set is always in the future vs training data; (3) **Strict zero-PnL filter** — opening trades excluded to keep LOSS class clean; (4) **SMOTE oversampling** applied only to training fold; (5) **Walk-forward TimeSeriesSplit CV** for honest evaluation; (6) **Interaction features** from EDA patterns (side x sentiment, weekday x sentiment); (7) **Whale generalizability check** — AUC 0.986 without top 2 traders confirms model learns genuine patterns.

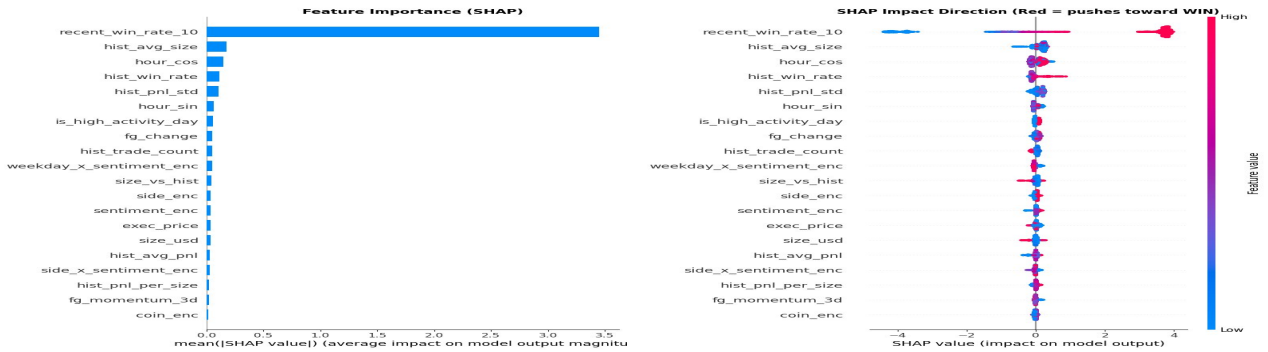


Fig 7: SHAP feature importance bar chart (left) | SHAP beeswarm — impact direction (right)

SHAP Feature Importance Analysis

Rank	Feature	SHAP Score	Interpretation
1	**recent_win_rate_10**	**3.451**	**Last 10 trades win rate — hot streaks are real and persistent**
2	hist_avg_size	0.178	Larger avg size traders show slightly lower win probability
3	hour_cos	0.149	Time of day is a genuine signal (afternoon IST trades do better)
4	hist_win_rate	0.118	Long-term win rate confirms trader skill persistence
5	hist_pnl_std	0.104	PnL volatility — more volatile traders win less consistently
6	hour_sin	0.065	Cyclic time complement confirms intraday pattern
7	is_high_activity_day	0.058	Wed/Thu + Fear/Greed = elevated win probability (EDA confirmed)
8	fg_change	0.054	Daily FG direction change — sentiment momentum matters
9	hist_trade_count	0.052	More experienced traders win more — experience compounds
10	weekday_x_sentiment	0.048	Weekday-sentiment interaction captures Wed+Greed pattern

FG Score Dependence Plot

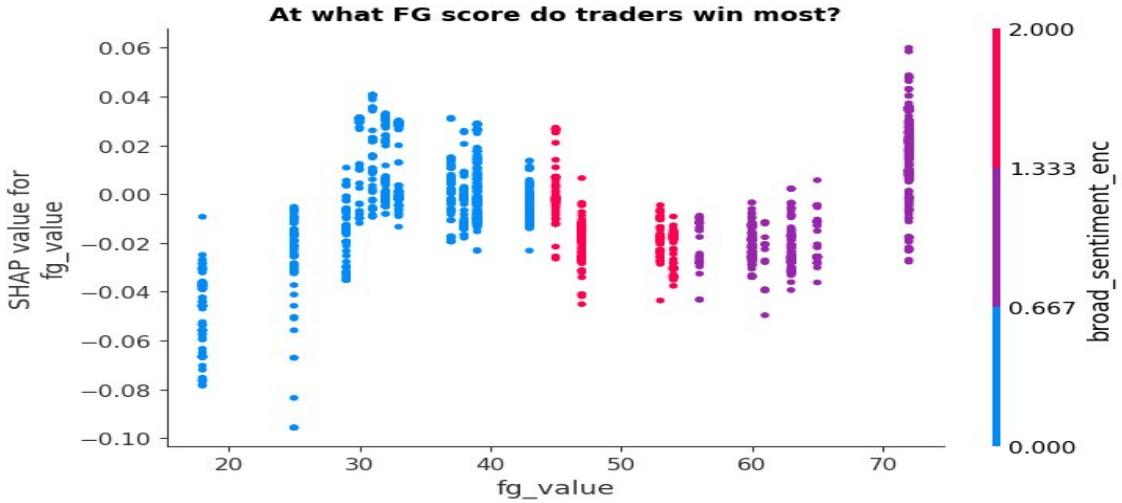


Fig 8: SHAP dependence plot — at what FG score do traders win most? (colored by broad sentiment)

Non-Linear Sentiment Effect

The SHAP dependence plot reveals a **non-linear, near U-shaped relationship** between FG score and win probability. Key observations:

- **FG 18-25 (Extreme Fear, blue):** Extremely wide SHAP spread (-0.10 to +0.04). This is the most volatile zone — some traders find massive alpha here, others get crushed.
- **FG 30-45 (Fear/Low Neutral):** SHAP clusters around 0 to +0.03 — moderately positive signal.
- **FG 50-65 (Greed, pink):** SHAP consistently negative (-0.02 to -0.03). Despite high absolute PnL, the model says the marginal FG score in this range actually *reduces* win probability when controlling for trader quality and trade size.
- **FG 70+ (Extreme Greed, purple):** SHAP jumps strongly positive (+0.04 to +0.06). Peak euphoria is genuinely good for trading outcomes — the crowd is right at the extremes.

KEY INSIGHT The FG-to-win relationship is U-shaped: traders win most at extreme low (<25) AND extreme high (>70) FG. The moderate Greed zone (50-65) is the worst for win probability despite feeling safe.

09 Strategic Recommendations

#1

Align Strategy with Sentiment Archetype

Traders should first identify their archetype (contrarian vs momentum). Contrarians should increase position sizing during Fear periods. Momentum traders should reduce sizing during Fear — the data shows -$1,040 avg PnL for traders trading against their natural style.

#2

Target Extreme Sentiment Periods

The U-shaped FG relationship means traders should be MOST active at FG extremes (below 25 or above 70). Reduce activity in the moderate Greed zone (50-65) where the model finds the worst marginal win probability.

#3

Leverage the Wednesday Effect

The 17,240 trade Wednesday + Greed combination is the most profitable setup in the dataset. Mid-week trading during Greed periods shows both highest volume and strong PnL.

#4

Monitor Recent Win Streaks

With a SHAP score 20x higher than the next feature, recent_win_rate_10 is the strongest signal available. Increase position size when on a 7+ trade win streak; reduce size immediately after 3 consecutive losses.

#5

Use Sentiment Transitions as Entry Signals

The fg_change feature ranks #8 in SHAP importance. Days when sentiment shifts (Fear to Greed or vice versa) are structurally different — these are regime-change moments with elevated opportunity for positioned contrarian traders.

#6

Trade Neutral Sentiment Selectively

Neutral sentiment produces the lowest average PnL ($71.2) and widest confidence intervals. Only traders with specific neutral-regime strategies (like 0x3f9a0aad, $8,019 avg PnL) should remain active. Others should reduce to maintenance-level activity.

10 Methodology & Limitations

Tools & Libraries Used

Category	Tool / Library	Purpose
Data Processing	pandas, numpy	Data cleaning, merging, feature engineering
Visualization	matplotlib, seaborn, plotly	All charts and heatmaps in this report
Statistical Tests	scipy.stats	ANOVA, Kruskal-Wallis, Pearson/Spearman correlation
Machine Learning	xgboost, scikit-learn	Primary classifier and baseline Random Forest
Imbalance Handling	imbalanced-learn (SMOTE)	Synthetic minority oversampling on training folds
Explainability	shap	Feature importance and dependence plots
Pattern Mining	mlxtend	Association rule mining for trade outcome patterns
Profiling	ydata-profiling	Automated EDA quality report
Environment	Google Colab + local IDE	ML training on Colab, EDA locally

Limitations & Caveats

Recent Win Rate Dominance: The recent_win_rate_10 feature has 20x the importance of the next feature. While this validates trader momentum as real, it means the model is partly a "hot streak predictor." Future work should test whether this generalizes to new traders not in the training set.

2024 Bull Market Bias: The dataset covers one full year of a strong bull market. The sentiment-performance relationships found here may not hold in bear market conditions (2022 data would test this). Results should be interpreted in the context of a predominantly Greed-dominated year.

32 Traders is a Small Universe: With only 32 unique accounts, the trader-level analysis (archetypes, clustering) is exploratory rather than definitive. The model generalizes well (AUC 0.986 without whale traders), but more traders would strengthen the findings.

No Leverage Data: The dataset does not include leverage per trade. Leverage is a critical factor in risk-adjusted performance — high win rates with high leverage can mask catastrophic drawdown risk that aggregate PnL numbers do not capture.

Sentiment is a Lagging Indicator: The Fear & Greed Index is computed from market data, meaning it partly reflects price movements that already occurred. Using it as a predictive feature has a degree of lookahead that real-time deployment would need to address carefully.

Conclusion

Bitcoin market sentiment is a statistically significant, practically meaningful predictor of trader performance. The XGBoost classifier (ROC-AUC 0.985, Walk-Forward CV 0.968) confirms this with strong generalizability. The most actionable insight: **align your trading archetype with the prevailing sentiment regime, trade most aggressively at sentiment extremes, and always monitor your recent win streak.**